From Analog Records to Computational Research Data: Building the AI-Ready Lab Notebook

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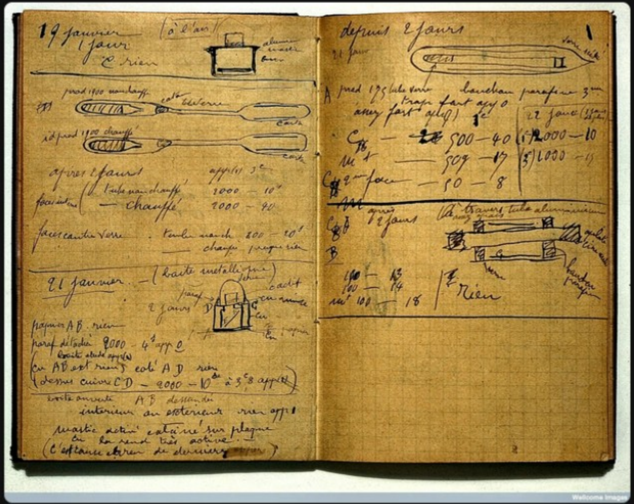
Institute for Data Driven Dynamical Design ID4

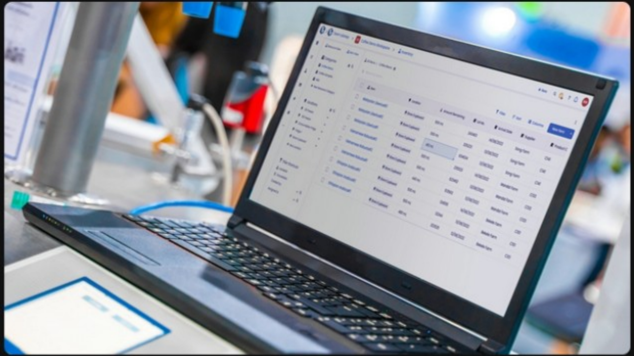
Reticular Synthesis & Materials Design Lab

Drexel University Metadata Research Center College of Computing & Informatics

Background

- Paper notebooks are an enduring fixture across laboratories
- Electronic lab notebooks (ELN's) changed scientific note taking, but cannot always be used
- Labs often accumulate stacks lab notebooks that remain completely un-digitized

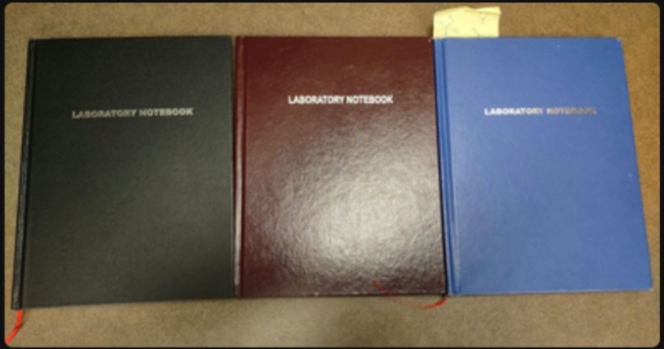




(1), (2)

Background

- Notebooks collections have potential to provide new insights into the successes, failures and pedagogy of experimental research
- Un-digitized notebooks become "data at risk"
- Research is needed to improve the ease of converting paper lab notebooks into digital, "AI-ready" archival records



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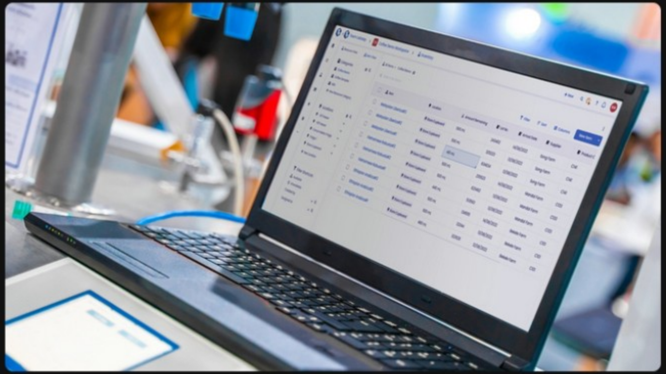


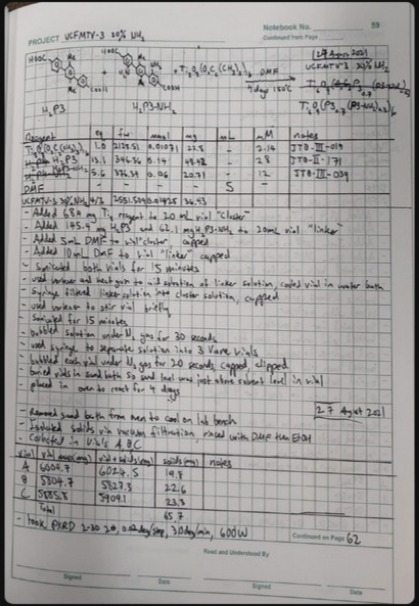
Goals

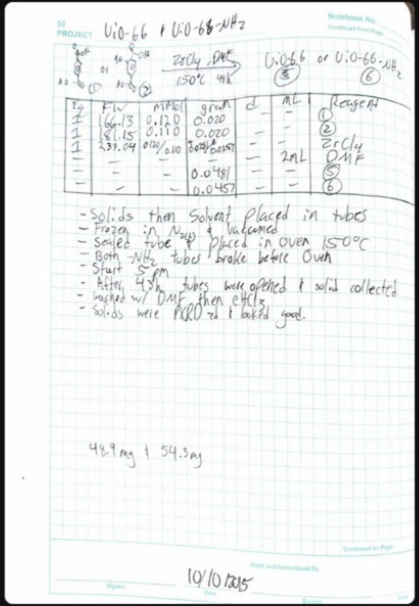
Overarching Goal: Advance methods for converting scanned handwritten notebooks into structured, FAIR, AI-ready data.

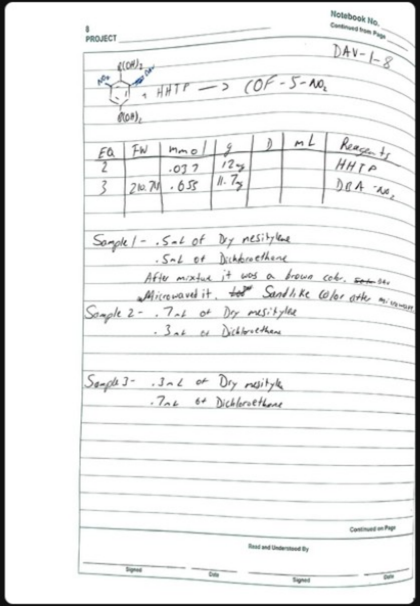
1. **Extraction**: Extract and archive data from notebooks to support computational analysis of these analog artifacts.
2. **Correction**: Identify common error patterns in extraction outputs and build a post-processing pipeline to correct them.
3. **Analysis**: Perform document classification and clustering analysis to answer scientific questions

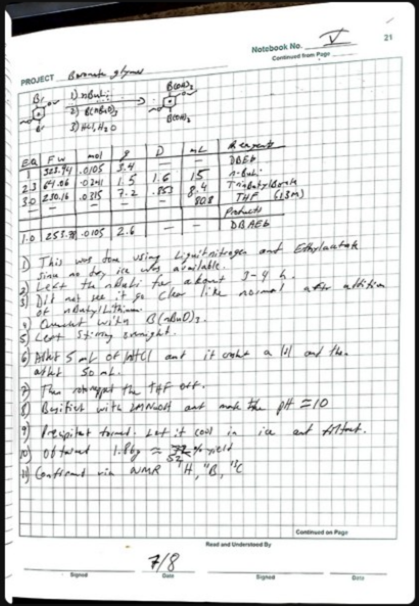
Data Repository

Collection of roughly 80 notebooks on **Metal Organic Framework** (MOF) crystal synthesis produced by the **Reticular Synthesis and Materials Design Lab** (RSMDL) at the University of Central Florida



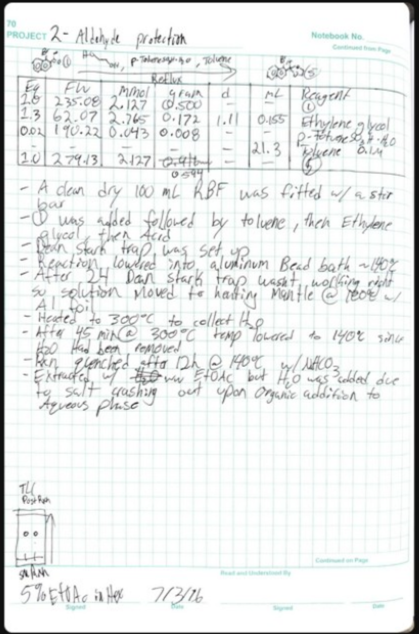






Methodology

1. Extract contents from pages via optical character recognition (OCR)
2. Post-process output to improve accuracy
3. Build an archive that can be presented by/in the style of an ELN
4. (Work in Progress) Further bolster the AI-readiness of the data by tokenizing protocols



Eq	FW	mmol	gram	d	mL	Reagent
1.0	235.08	2.127	0.500	-	-	①
1.3	62.07	2.765	0.172	1.11	0.155	Ethylene glycol
0.02	190.22	0.043	0.008	-	-	p-TolueneSO3H·H2O
-	-	-	-	-	21.3	Toluene 0.1M
1.0	279.13	2.127	~~0.41~~ 0.594	-	-	⑤

```
- A clean 100 mL dry round-bottom flask (RBF) was fitted with a stirrer.
- Q was added, followed by toluene, then ethylene glycol, and then acid.
- A Dean-Stark apparatus was set up.
- The reaction was lowered into an aluminum bead bath at approximately 140°C.
- After 24 hours, the Dean-Stark trap was checked; it was working correctly,
  so the solution was moved to a heating mantle at 120°C with aluminum foil.
- Heated to 300°C to collect H20.
- After 45 minutes at 300°C, the temperature was lowered to 140°C
  since H20 had been removed.
- The reaction was quenched after 12 hours at 140°C with NHCO2.
- Extracted with EtOAc, but H20 was added due to salt crashing out upon
  organic addition to the aqueous phase.
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1	325.74	.06140	.02	–	–	HHTP
1.7	237.71	.1044	.025	–	–	tetrafluoro diboranic acid.
/	/	/	/	/	/	Product
1	–	.06140	.04	–	–	COF-5-F_4
					2.0	1:1 DCE/Dioxane .03M

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,EQ,FW,mmol,/2,D /,ML,Reagents
0,1.0,325.74,.06/40,0.02,-,-,HHTP
1,1.7,237.71,. 1044,",025",-,-,tetrafluoro diboranic acid.
2,,,/ /,,,,Product
3,1.0,,. 06140,",04",-,-,COF-5-F
4,,,,",04",,2,1:1 DIE/Dioxone . 03M
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Reagents	EQ	FW	Mmol	G	D	M L
HHTP	1	325.74	0.0614	0.02	-	-
tetrafluo...	1.7	237.71	0.1044	0.025	-	-
Product	-	-	-	-	-	-
COF-5-F	1.0	-	.06140	.04	-	-
1:1 DIE/...	-	-	-	.04	-	2

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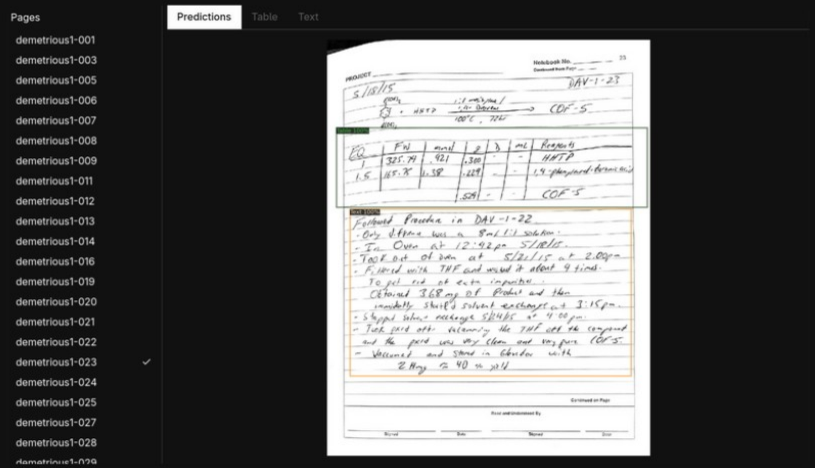

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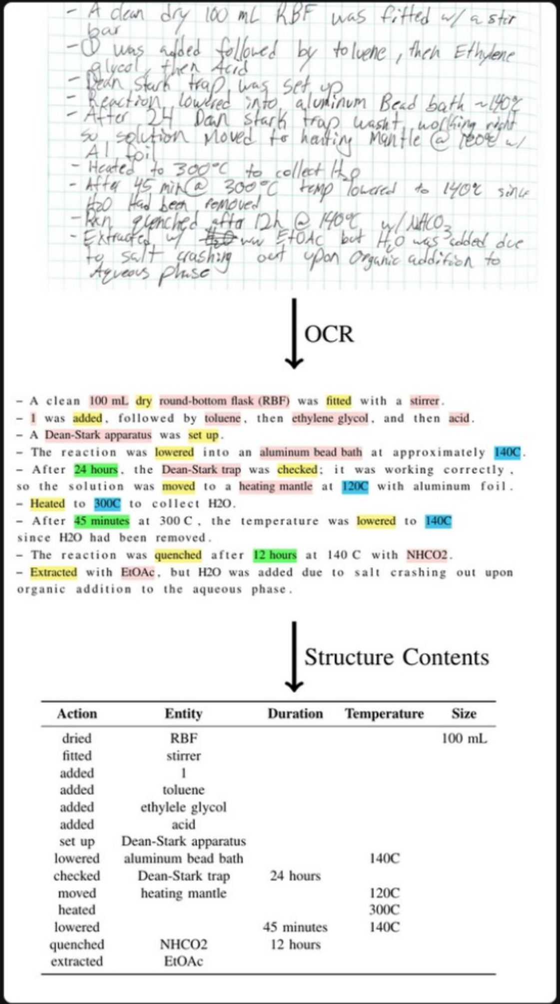


Action	Entity	Duration	Temperature	Size
dried	RBF			100 mL
fitted	stirrer			
added	1			
added	toluene			
added	ethylele glycol			
added	acid			
set up	Dean-Stark apparatus			
lowered	aluminum bead bath		140C	
checked	Dean-Stark trap	24 hours		
moved	heating mantle		120C	
heated			300C	
lowered		45 minutes	140C	
quenched	NHCO2	12 hours		
extracted	EtOAc			

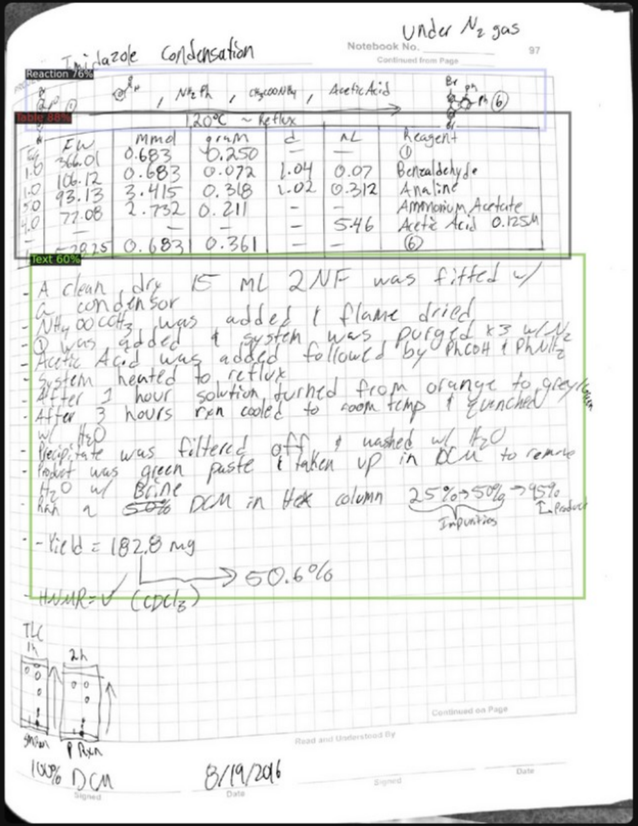
Content Extraction

Recently, the capabilities and performance of document content extraction software have rapidly increased

(3)

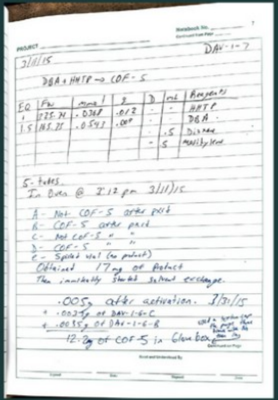
Content Extraction

Two step Detectron2 and HandwritingOCR method



olmOCR 2

Page 5 490 tokens processed Copy



3/11/15

DAV-1-7

DQA + HHTP → COF-S

EQ	MW	mmol	g	D mL	Reagents
1	325.74	0.0368	0.12	-	HHTP
1.5	165.75	0.543	0.09	-	DQA
				.5	Dioxane
				.5	Mesitylene

5-tubes. In Oven @ 3:12 pm 3/11/15

A - Not COF-S after px1t B - COF-S after px1t C - Not COF-S " D - COF-S " E - Spilled vial (no product)

Obtained 17 mg of Product Then immediately started Solvent exchange.

.005g after activation. 3/31/15

- .0037g of DAV-1-6-C
- .0035g of DAV-1-6-B 12.2mg of COF-S in glove box

Post Processing

- Regardless of the extraction method, there are always some errors.
- Checks and fixes for tables in RSMDL notebooks include:
  - Removing any empty rows/columns and realigning elements.
  - Correcting header errors (e.g., reading "FW" as "MW").
  - Validating that numerical columns contain only numbers.
  - Correctly handling slashed/blacked out cells
  - Within rows: $\mathrm{FW} * \mathrm{mmol} = \mathrm{grams}$
  - Between columns: $\mathrm{mmol}_i/\mathrm{Eq}_i = \mathrm{mmol}_j/\mathrm{Eq}_j$ for rows i and j.

EQ	FW	mmol	g	D	mL	Reagents
1	325.74	.06140	.02	–	–	HHTP
1.7	237.71	.1044	.025	–	–	tetrafluoro diboranic acid.
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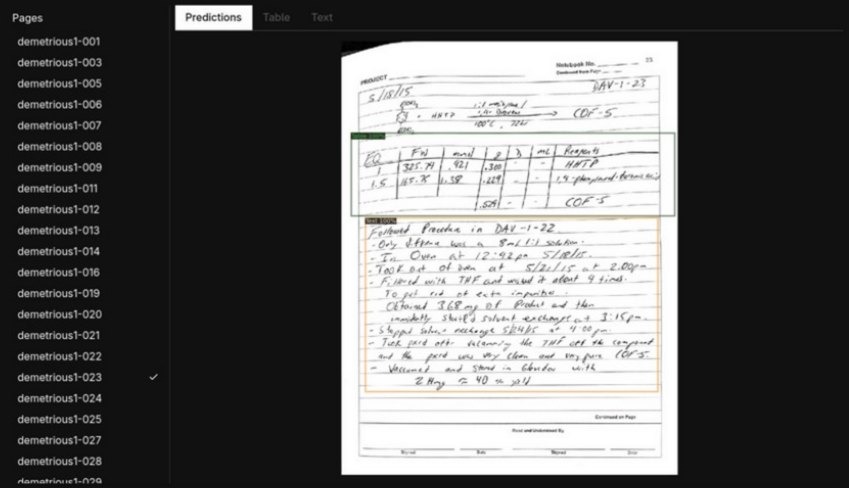
COF-5-F	1.0	-	.06140	.04	-	-
1:1 DIE/...	-	-	-	.04	-	2

Archival and Presentation

- Following extraction and post-processing, an archive of the contents is generated
- Archive can be rendered in ELN-like web interface, or potentially imported into full fledged ELN in the future



Results

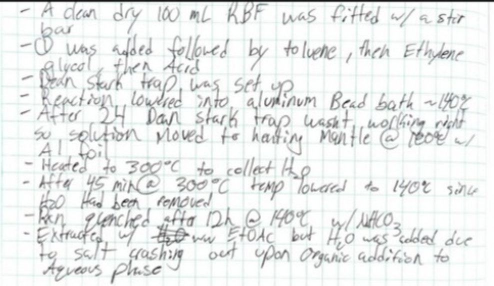
Technique	Mistakes	Error Rate	Improvement Rate
D2+HOCR*	199	18.22%	-
olmOCR 2	61	5.59%	-
D2+HOCR Post-Processed	62	4.76%	73.87%
olmOCR 2 Post-Processed	46	4.21%	24.69%

30 page sampling contains **1,092** cells

*Detectron2 and HandwritingOCR

Future Work

- Now working on converting the contents of text blocks into a tokenized list of actions
- Work is an extension of previous work by our group using LLMs to tokenize methodology sections from MOF research papers (4)
- Tokenizing protocols can greatly improve ML analyses of their content (5), increasing data's "AI-readiness" and the potential for knowledge discovery



OCR

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Conclusion

- Overall Goals:
  - Make data contained in lab notebooks AI-ready
  - Mine information in notebooks for scientific insights
- Content extraction aspects of work largely completed
- Work to tokenize and further process digital notebook archives to bolster AI-readiness ongoing, moving towards using notebooks to answer scientific questions

- 1. Cliff Pickover [@pickover]. Physics. Marie Curie's (1867–1934) century-old experimental notebook [Internet]. Twitter. 2022 [cited 2025 Dec 4]. Available from: https://x.com/pickover/status/1526948489099464704 (2)
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- 4. Zhao X, Rojas JFF, Furst J, Ardila K, Langlois K, An Y, et al. Expert-Guided LLM Approach for Sequence-Aware Extraction of MOF Synthesis [Internet]. ChemRxiv; 2025 [cited 2025 Nov 23]. Available from: https://chemrxiv.org/engage/chemrxiv/article-details/689c179c728bf9025e39cb58 (12)
- 5. Li A, Wang X, Wang W, Zhang A, Li B. A survey of relation extraction of knowledge graphs. In Springer; 2019. p. 52–66. (12)

Thank You!

Questions?

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